

Cloud Computing Tutorial 4 & 5

CloudSim Simulation

Observation:

Tutorial 4 & 5:

cloudsim simulation

• cloudsim

→ cloudsim is a simulation tool that helps researchers and students to study a how cloud computing works without requiring any real cloud infrastructure. Its applications include to test and model cloud systems, virtual machines, data centres, scheduling policies and deploying and maintaining them.

① What does cloudsim do?

- Evaluate performance metrics.
- Optimizing cloud services
- Simulate cloud infrastructure
- Testing of scheduling & Resource allocation.

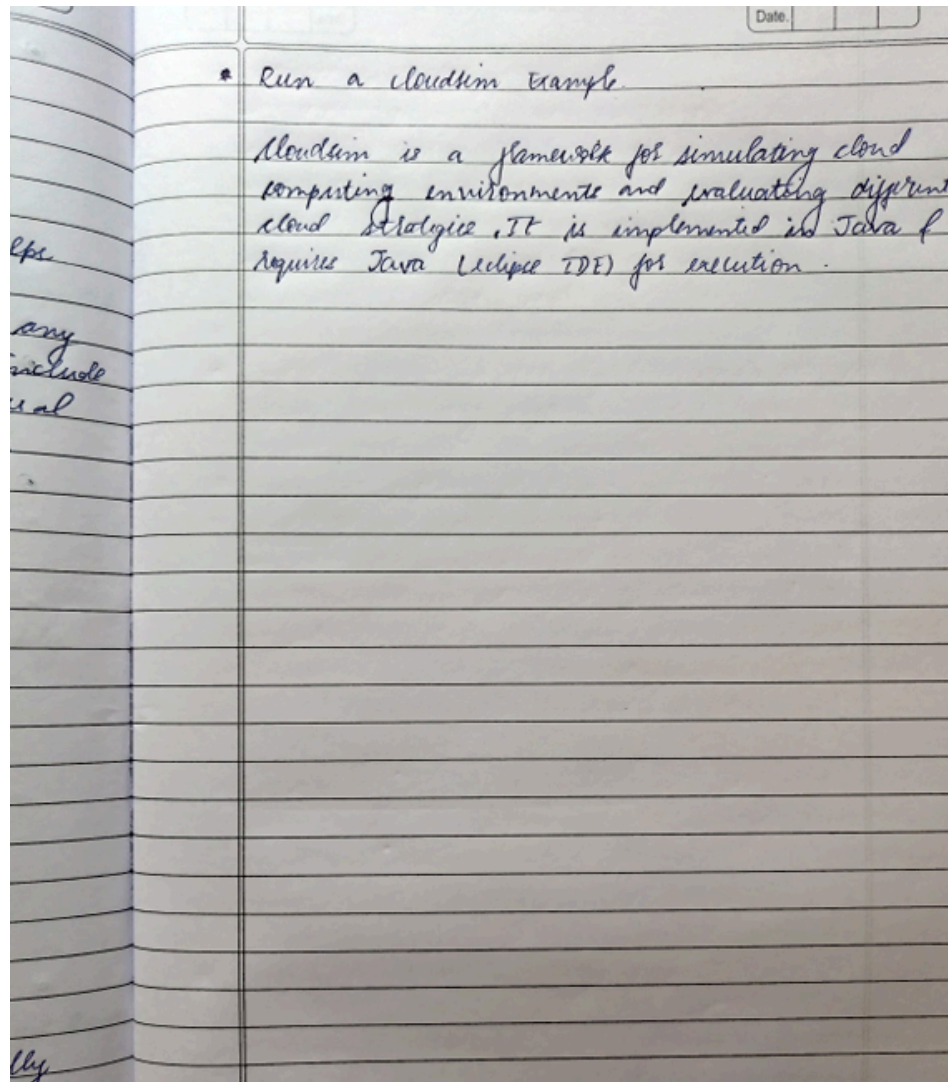
② Steps to be followed.

* Prerequisites:

- JDK 8 or greater.
- Cloudsim source code
- Eclipse IDE
- Apache Commons Math library.

* Add Cloudsim source code to eclipse, manually copy & paste the code into eclipse project.

* Add required libraries in the form of external JAR files.



CloudSim demonstration:

```
C:\Users\Admin>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\Admin>
```

Fig 1.1 JDK installation verification

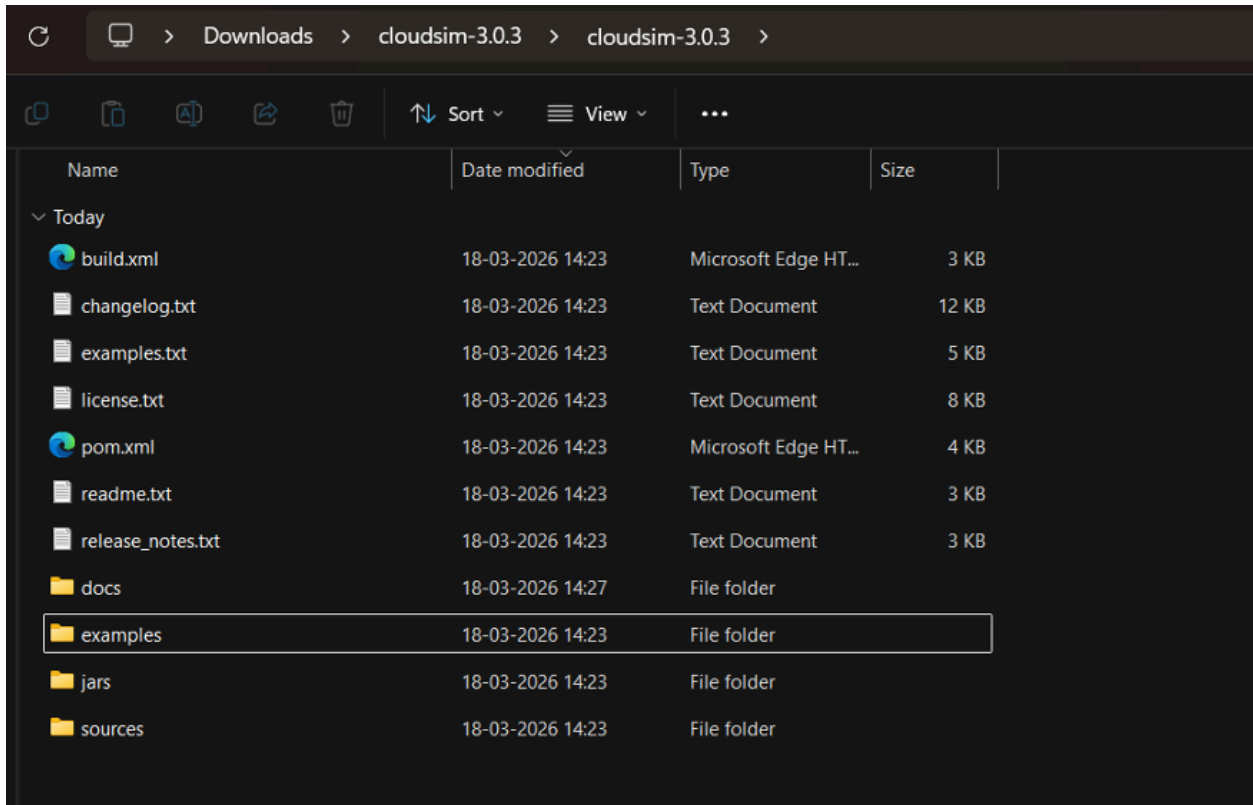


Fig 1.2 Cloudsim zip folder extraction

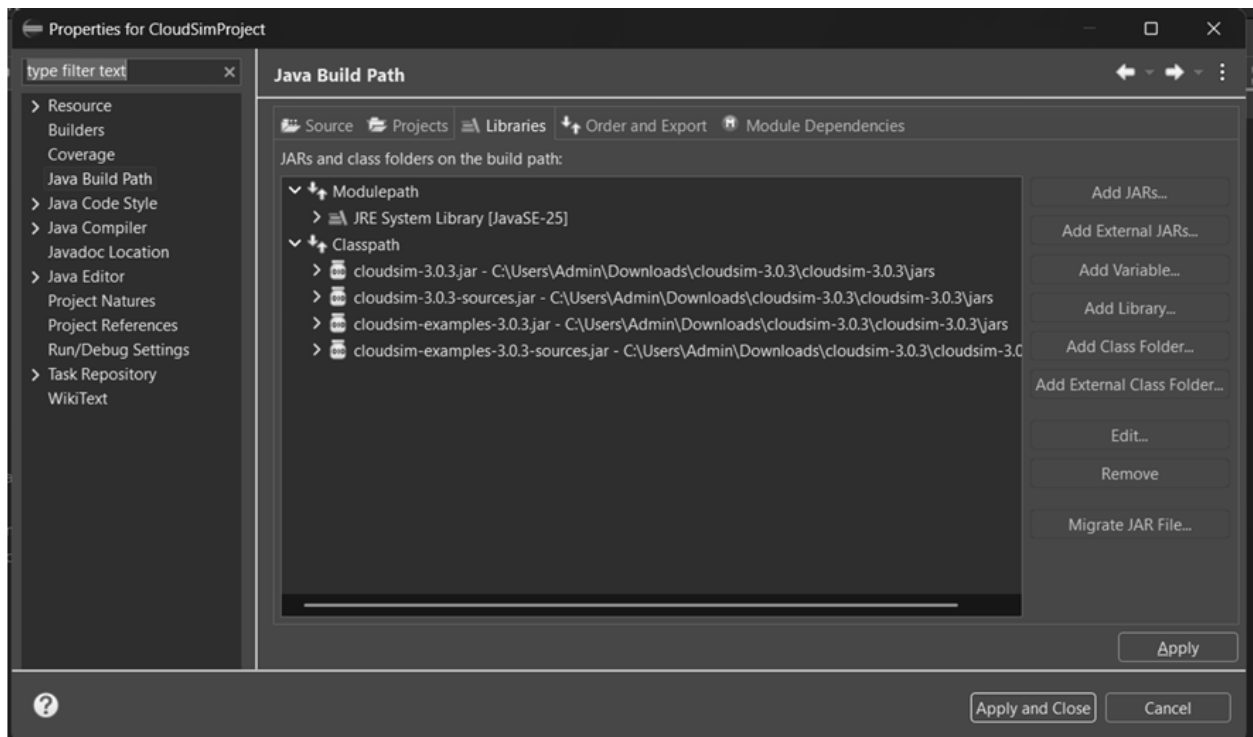


Fig 1.3 CloudSimProject properties

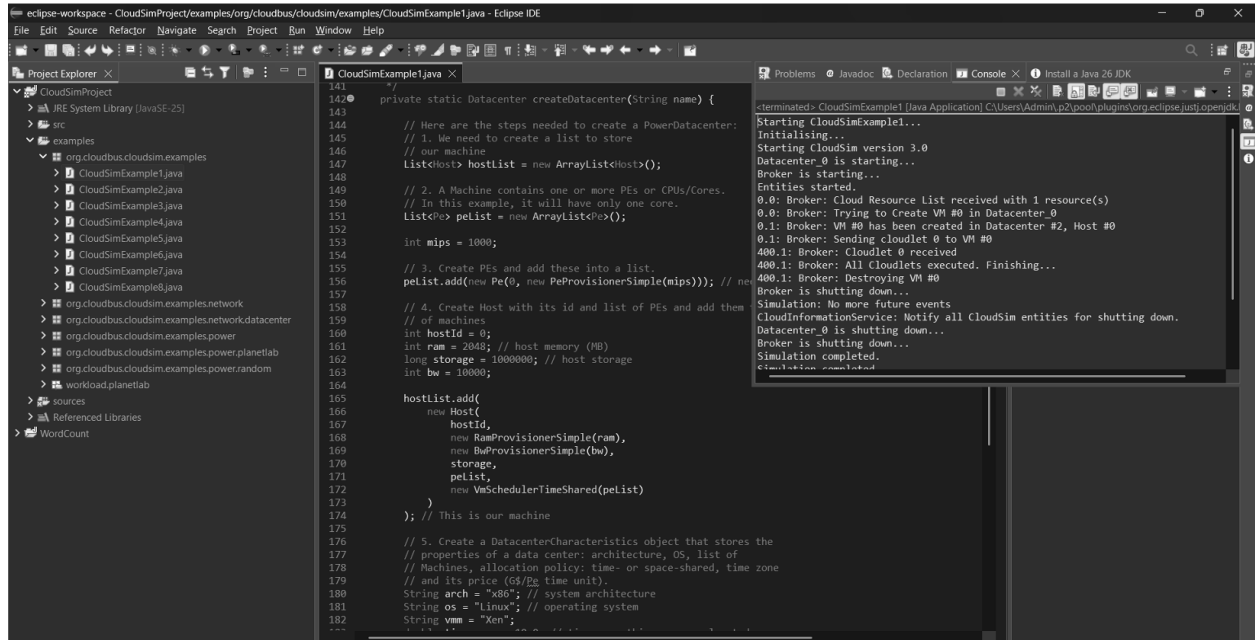


Fig 1.4 Running CloudSimProject - CloudSimExample1.java program

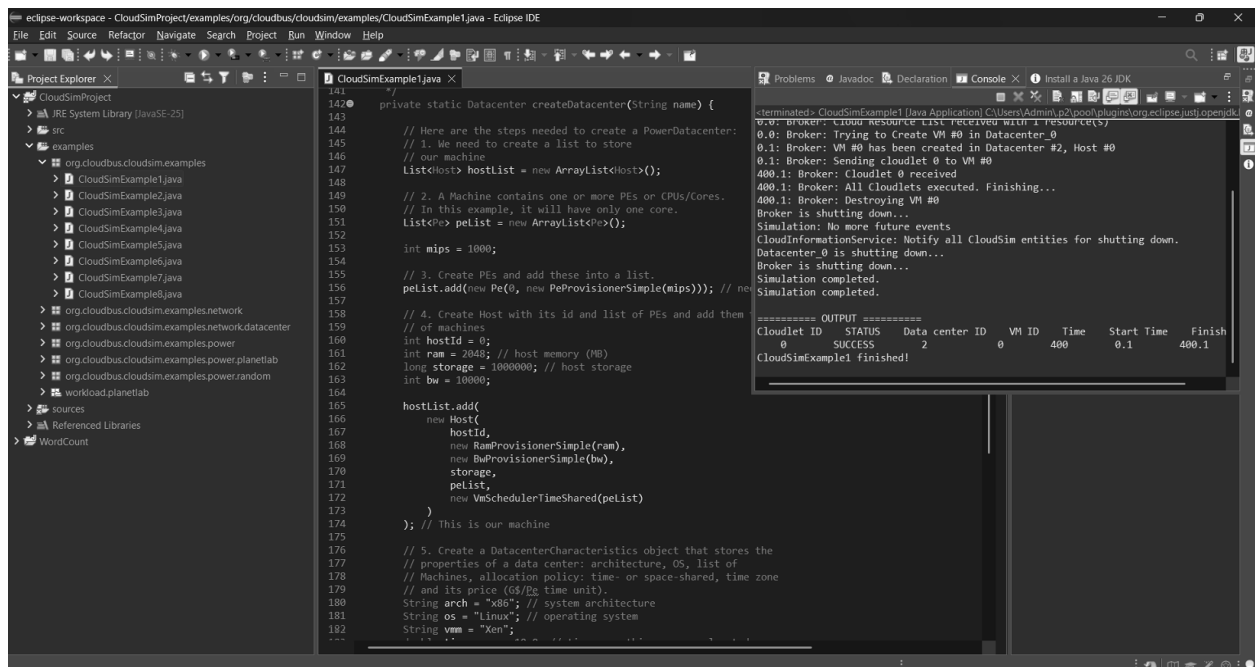


Fig 1.5 Simulation successful